<u>SML Assignment 5</u>

Implementation of K-Nearest Neighbors (KNN) for Classification and Regression

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Aim: To implement the K-Nearest Neighbors (KNN) algorithm for classification and regression tasks and analyze its performance on real datasets.

Objective: To understand the working of the K-Nearest Neighbors algorithm:

- To implement KNN for classification and regression problems
- To evaluate model performance using suitable metrics
- To analyze the impact of different values of K on prediction accuracy
- To visualize and interpret the results

Problem statement: Use the given dataset(s) to demonstrate the application of K-Nearest Neighbors (KNN) classification. The task is to identify the class of the query instance using k-nearest neighbors.

Theory:

K-Nearest Neighbours:

K-Nearest Neighbors (KNN) is a supervised machine learning algorithm used for both classification and regression tasks. It works on the principle that similar data points exist in close proximity in the feature space. For a new query instance, the algorithm finds the K closest data points from the training set and determines the output based on majority voting (classification) or averaging (regression).

2. Mathematical Formula:

The distance between two points is usually computed using Euclidean Distance:

$$d(x, y) = \sqrt{\sum_{i=1}^{n} (y_i - x_i)^2}$$

Where:

- Xi= feature value of query point
- Yi = feature value of training point
- n = number of features

For classification:

Predicted Label=mode of nearest K labels

For Regression:

$$\hat{y} = \frac{1}{K} \sum_{i=1}^{K} y_i$$

3. Importance of KNN

- Simple and intuitive algorithm
- Works well for nonlinear data
- Requires no explicit training phase
- Effective for small to medium datasets
- Can be used for both classification and regression

4. Applications of KNN

- Medical diagnosis
- Fraud detection
- Recommendation systems
- Image recognition
- Pattern recognition
- Text classification

5. Performance Metrics

Accuracy: Measures proportion of correctly classified samples.

Precision: Measures correctness of positive predictions.

Recall: Measures ability to detect actual positives.

F1-Score: Harmonic mean of precision and recall.

Confusion Matrix: Shows TP, TN, FP, FN values.

Cross-Validation: Technique used to evaluate model performance by splitting data into multiple folds.

6. Dataset Description

Dataset Used: Heart Failure Clinical Records

- Dependent variable: DEATH_EVENT
- Independent variables: age, ejection fraction, serum creatinine, platelets, blood pressure, diabetes, smoking, etc.
- Dataset contains numerical medical attributes for patient survival prediction

Dataset Used: Bike Sharing Dataset

- Dependent variable: cnt (total bike rentals)
- Independent variables: season, year, month, holiday, weekday, working day, weather, temperature, humidity, wind speed
- Dataset contains environmental and temporal attributes for rental demand prediction

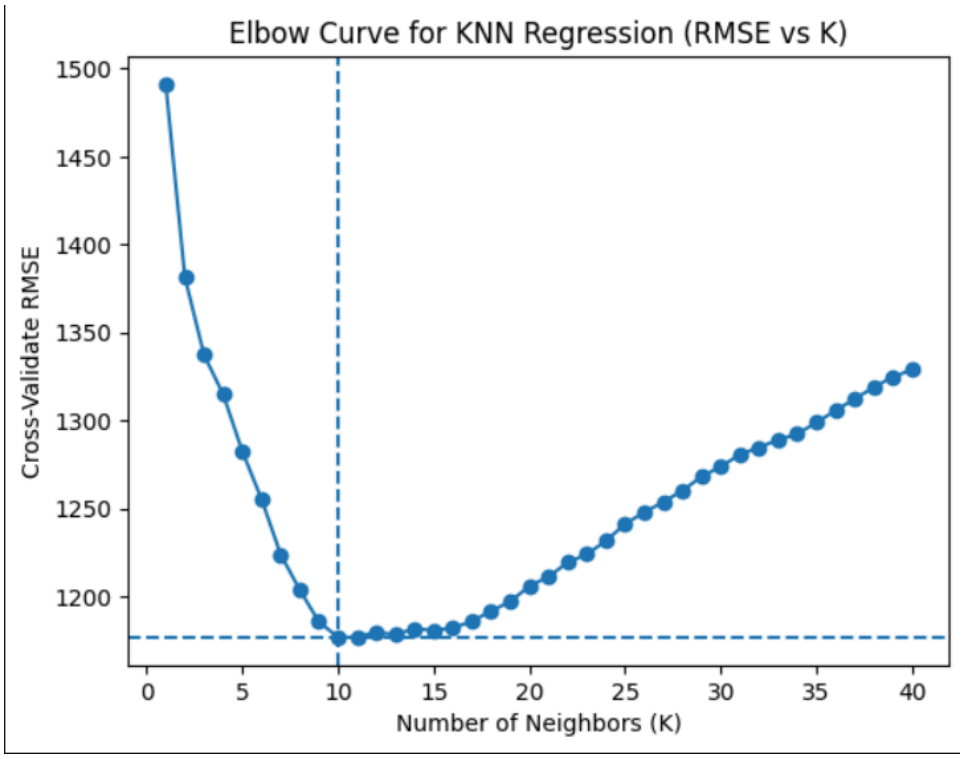
Methodology

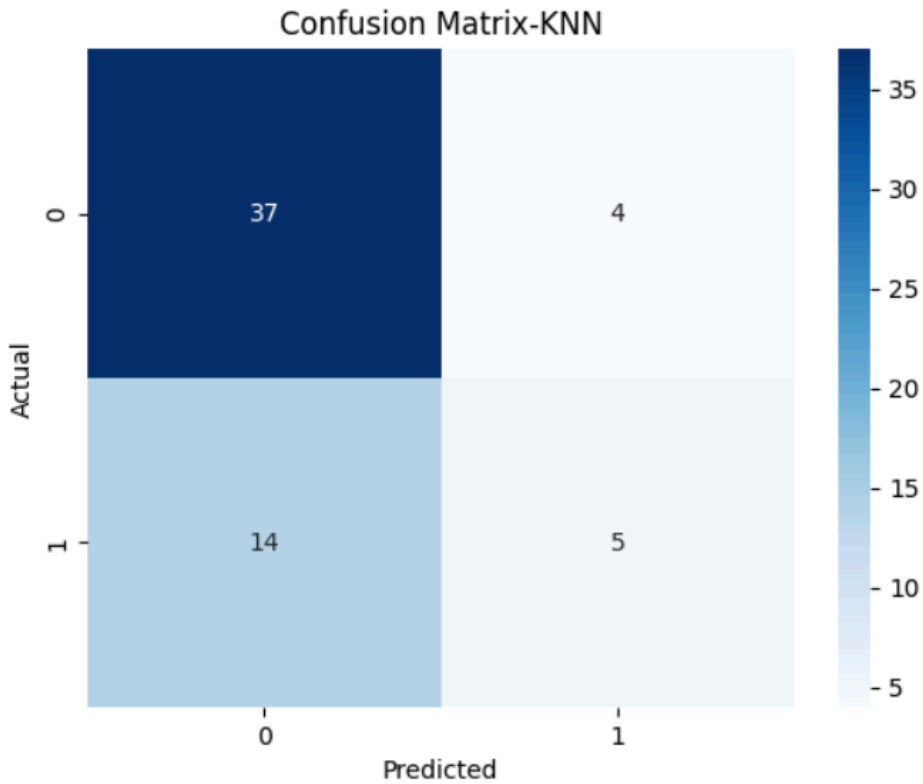
Input:

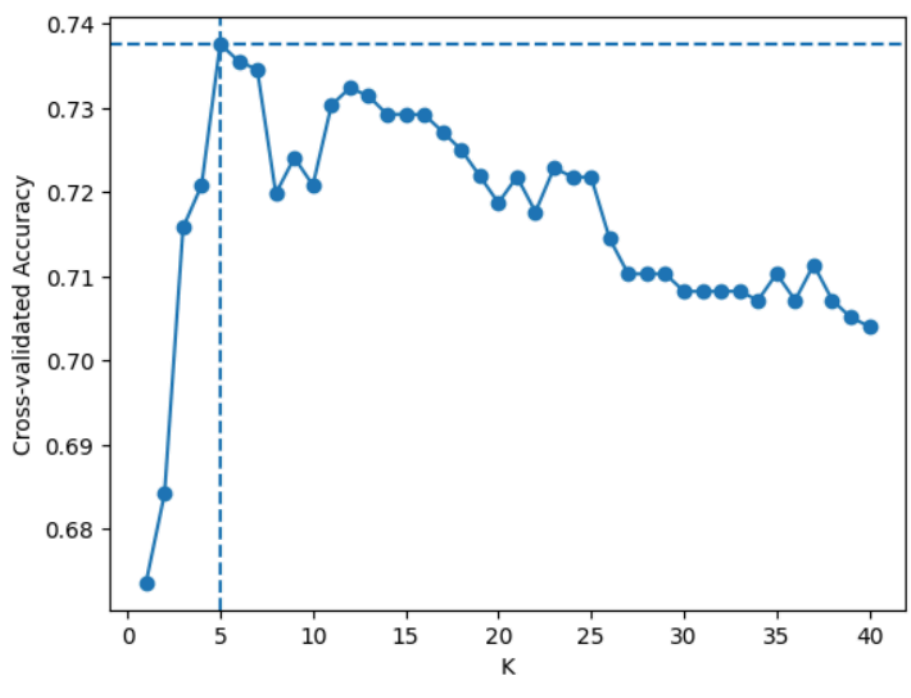
- Dataset containing features and target variable
- Value of K selected through cross-validation

Output:

- Predicted class labels (classification)
- Predicted numerical values (regression)
- Accuracy, confusion matrix, F1-score, RMSE, R^2
- Graphs showing performance vs K




Confusion Matrix-KNN
37
4
14
5
Actual
0
1
Predicted
0
1
35
30
25
20
15
10
5


Cross-validated Accuracy
0.74
0.73
0.72
0.71
0.70
0.69
0.68
0
5
10
15
20
25
30
35
40
K

Conclusion

Impact of Small K

- Low bias
- High variance
- Sensitive to noise
- May overfit

Impact of Large K

- High bias
- Low variance
- Smoother decision boundary
- May underfit